TECHNICAL MANUAL & PROXY DRAWINGS

Model: HP-50T-PRO (50-Ton Hydraulic Press)

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Strictly for Training and Simulation Purposes

1. General Information

This manual provides the proxy operational parameters, mechanical layout, and electrical schematics for the **HP-50T-PRO 50-Ton Hydraulic Press**. This equipment is designed for heavy-duty industrial applications including metal forming, deep drawing, blanking, and precision straightening.

1.1 Specifications

Parameter	Specification
Maximum Pressing Force	50 Metric Tons
Cylinder Stroke	300 mm
System Operating Pressure	250 Bar (3625 PSI)
Main Motor Power	7.5 kW (10 HP)
Power Supply Requirements	400V / 3-Phase / 50Hz
Machine Weight	1,850 kg

DANGER: HIGH PRESSURE SYSTEM

This machine operates using highly pressurized hydraulic fluid. Never attempt to loosen, tighten, or modify hydraulic lines while the system is powered or retaining residual pressure. Doing so may result in fluid injection injuries or fatal accidents.

2. Safety Protocols

All operators must adhere to the following safety guidelines prior to engaging the hydraulic press:

- **Pre-Operation Inspection:** Ensure the optical light curtains are functional and free from obstructions. Test the Emergency Stop (E-Stop) buttons located on the main panel and the foot pedal assembly.

- **PPE Requirements:** Safety goggles, steel-toed boots, and heavy-duty industrial gloves are mandatory.
- **Lockout/Tagout (LOTO):** Before commencing any maintenance, the main electrical isolator must be switched off and padlocked. Residual hydraulic pressure must be bled via the bypass valve.

3. Proxy Mechanical Drawings

The following schematic illustrates the primary structural and operational components of the HP-50T-PRO hydraulic press. This diagram serves as a proxy representation for spatial orientation and component identification.

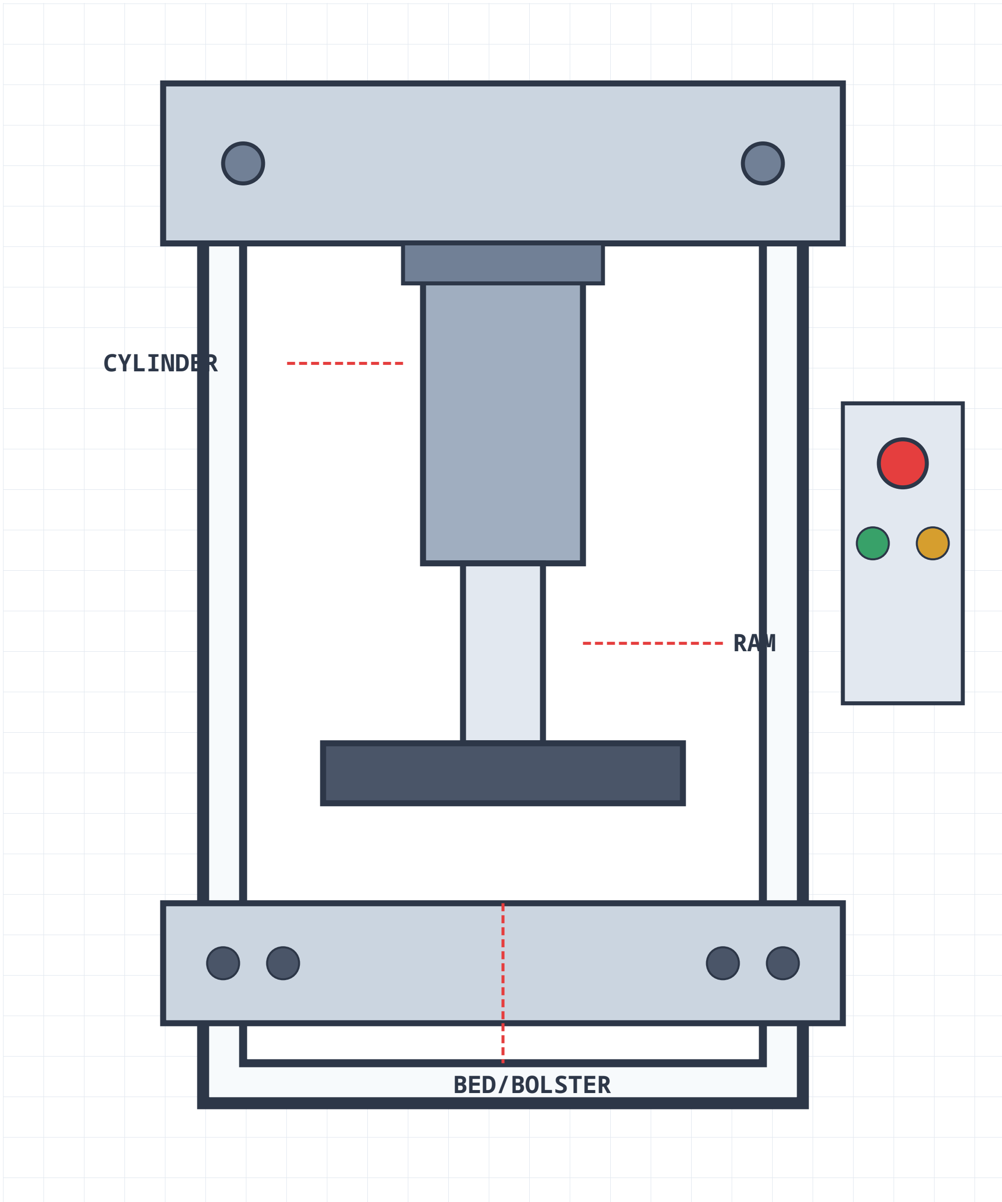

Figure 3.1: Front Elevation Mechanical Schematic (H-Frame Architecture)

4. Proxy Electrical Drawings

The simplified control circuit diagram below demonstrates the safety interlocks and main motor activation logic. The system utilizes a dual-channel safety relay connected to the E-Stop and Light Curtains.

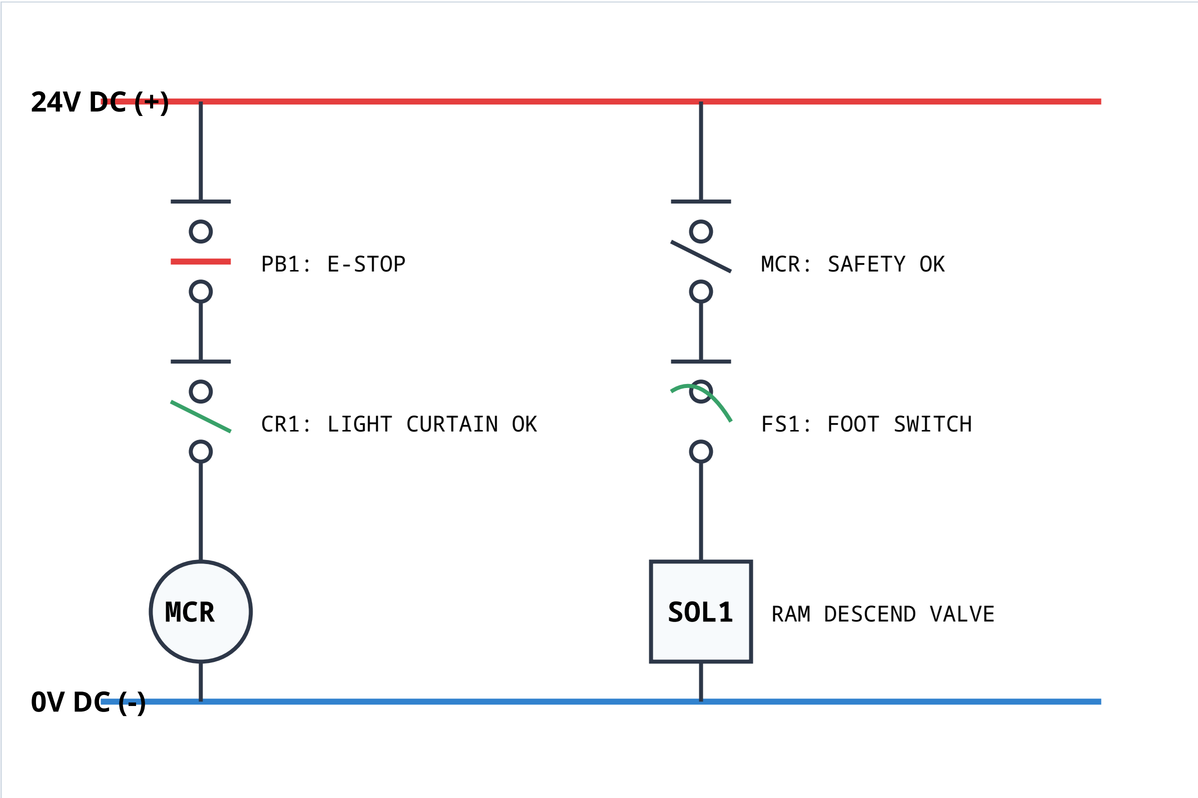

Figure 4.1: Control Logic Schematic (Simplified Ladder Logic)

5. Operating Instructions

Standard operating procedure requires a two-step validation sequence before the pressing cycle can begin.

1. **Power Up:** Turn the main rotary disconnect switch to the "ON" position. Verify that the 24V DC power indicator is illuminated on the control panel.
2. **System Reset:** Clear the emergency stop by twisting the red E-Stop button clockwise. Press the blue "RESET" button to energize the Master Control Relay (MCR).
3. **Setup Mode:** Select "JOG" mode on the HMI. Use the dual hand controls to slowly lower the ram to the desired limit to set the mechanical limit switches.

4. **Production Cycle:** Switch to "AUTO" mode. Insert the workpiece, step clear of the light curtains, and depress the foot switch to initiate the hydraulic cycle.

6. Routine Maintenance Schedule

All maintenance activities must be logged in the Machine History database. Use genuine ISO VG-46 hydraulic oil for top-ups.

Interval	Task Description
Daily (Pre-Shift)	Inspect hydraulic hoses for weeping/leaks. Wipe down optical light curtains.
Weekly	Check hydraulic oil level in the reservoir sight glass. Lubricate ram guide rails with lithium grease.
Monthly	Inspect structural bolts on the crosshead for correct torque. Clean the motor cooling fins.
Bi-Annually (2000 Hrs)	Replace hydraulic oil return filter. Analyze oil for particulate contamination.